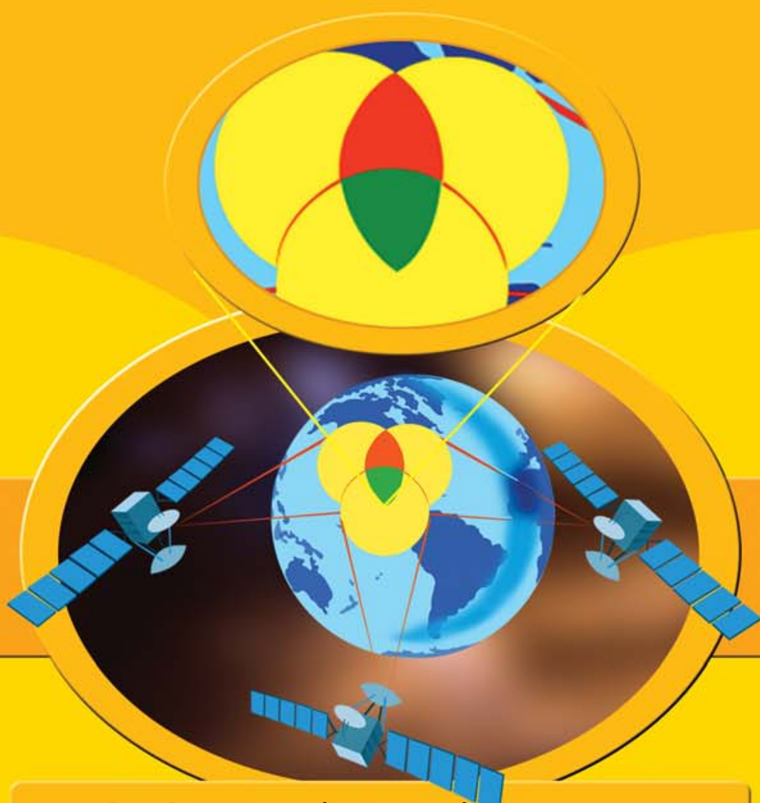
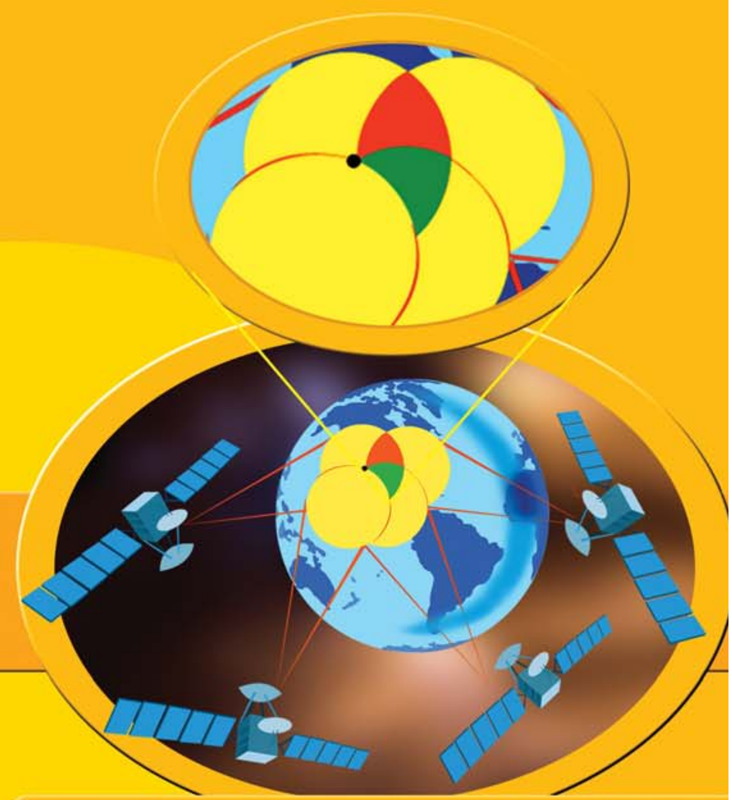
How GPS Works





III. The Three-Satellite Scenario

Now, when a third satellite is involved, the receiver knows it is within the green area (as in illustration above)—the intersection of three circles. Ideally, this green area should be a single point, but in the real world, it is not. A fourth satellite is therefore required to pinpoint the location of the receiver.

IV. The Fourth Satellite To The Rescue

When there is information from a fourth satellite, the receiver knows exactly where it is in three dimensions. The latitude and longitude as well as the altitude are then known. The fourth satellite's information does two things; refine what the receiver already knows, and pinpoint altitude adding to the accuracy.

Estimating Distance

One question has remained unanswered thus far: how does the GPS receiver know the distance between itself and a satellite? How can it estimate this?

The answer is that each GPS satellite is constantly sending out a unique stream of 0s and 1s, which identify the satellite. Each GPS receiver is synchronised with all the satellites, as mentioned earlier, and it has the same stream of 0s and 1s generated within it, in synchrony. When a receiver gets a certain signal from a satellite, it guages the time shift between the internal sequence and that of the satellite. Thus, figuring out how long it has taken the signal to reach it is possible. With this time gap known, the distance can be figured, since radio waves travel at the speed of light.

Differential GPS

So far, that is, with four satellites in the picture, the precision of pointing out a location is between 10 and 30 metres. To refine this further and improve accuracy to within a couple of metres, differential GPS (DGPS) is used.

A GPS receiver, known as the base station, constructed at a place where the latitude, longitude and altitude are precisely known, through geographical surveys, is used to send out DGPS signals. The receiver determines its position according to the information received from three satellites. This recorded position is bound to be a little different from the known position, and it is this difference that the unit broadcasts, via its antenna. Using the difference in signal in addition to the information the receiver already has from the satellites, other GPS receivers can refine their position.

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